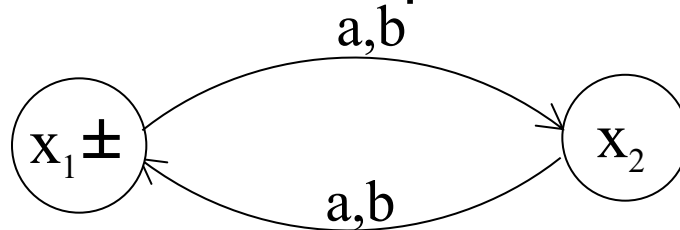


RECAP Lecture 13

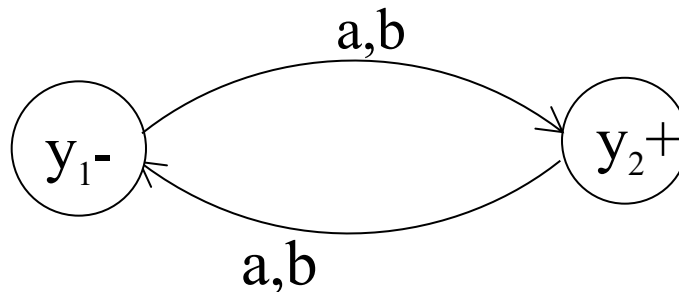
- Examples of Kleene's theorem part III
(method 1) continued ,Kleene's theorem part III
(method 2: Concatenation of FAs),
- Example of Kleene's theorem part III
(method 2 : Concatenation of FAs)

Example

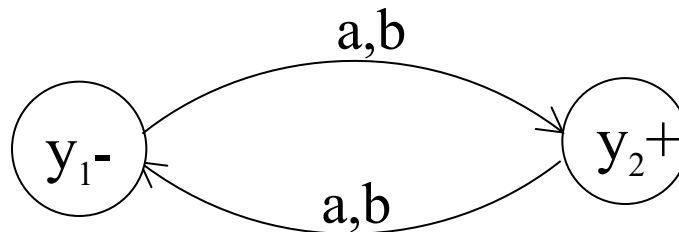
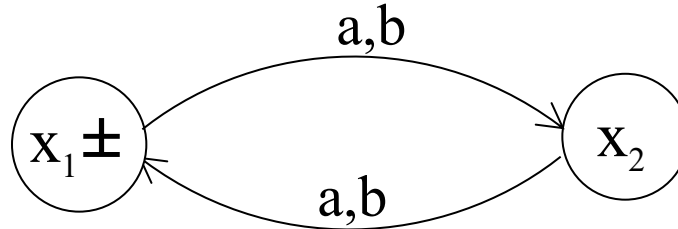
Let $r_1 = ((a+b)(a+b))^*$ and the corresponding FA_1 be



also $r_2 = (a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$ and FA_2 be



Example continued ...

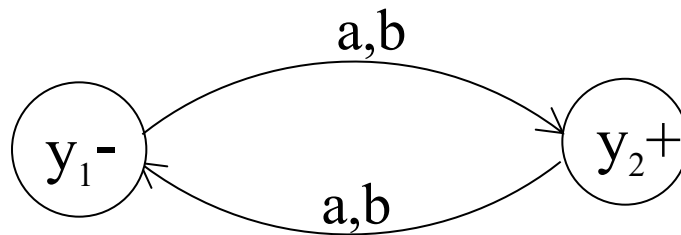


Old States	New States after reading	
	a	b
$z_1 - \text{?}(x_1, y_1)$	$(x_2, y_2) \text{ ? } z_2$	$(x_2, y_2) \text{ ? } z_2$

Example continued ...

Old States	New States after reading	
	a	b
$z_2 + \neg(x_2, y_2)$	$(x_1, y_1) \neg z_1$	$(x_1, y_1) \neg z_1$

Example continued ...



Kleene's Theorem Part III

Continued ...

- **Method3: (Closure of an FA)**

Building an FA corresponding to r^* , using the FA corresponding to r .

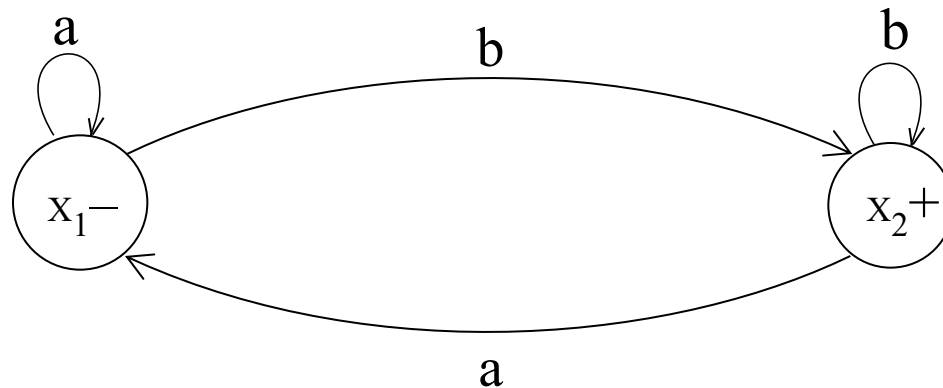
It is to be noted that if the given FA already accepts the language expressed by the closure of certain RE, then the given FA is the required FA. However the method, in other cases, can be developed considering the following examples

Closure of FA Continued ...

Closure of an FA, is same as concatenation of an FA with itself, except that the initial state of the required FA is a final state as well. Here the initial state of given FA, corresponds to the initial state of required FA and a non final state of the required FA as well.

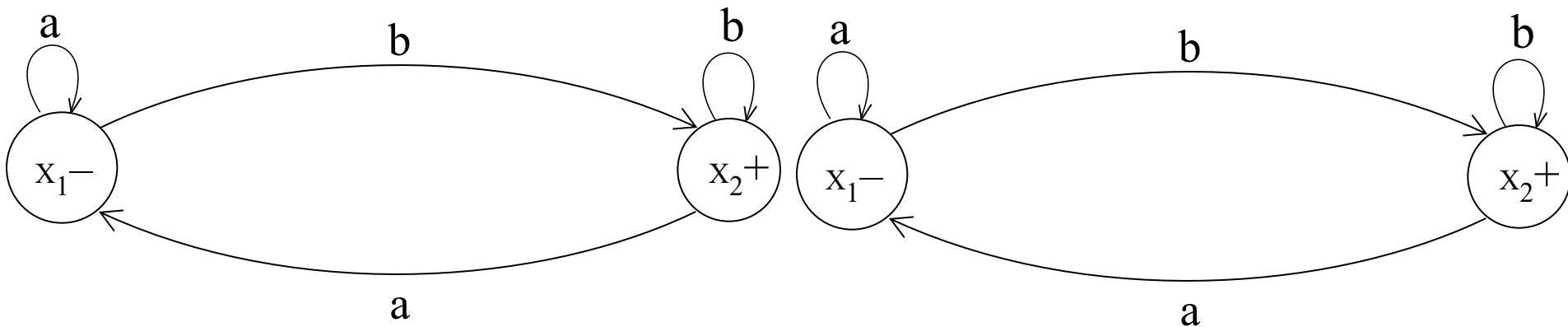
Example

Let $r=(a+b)^*b$ and the corresponding FA be



then the FA corresponding to r^* may be determined as under

Example continued ...



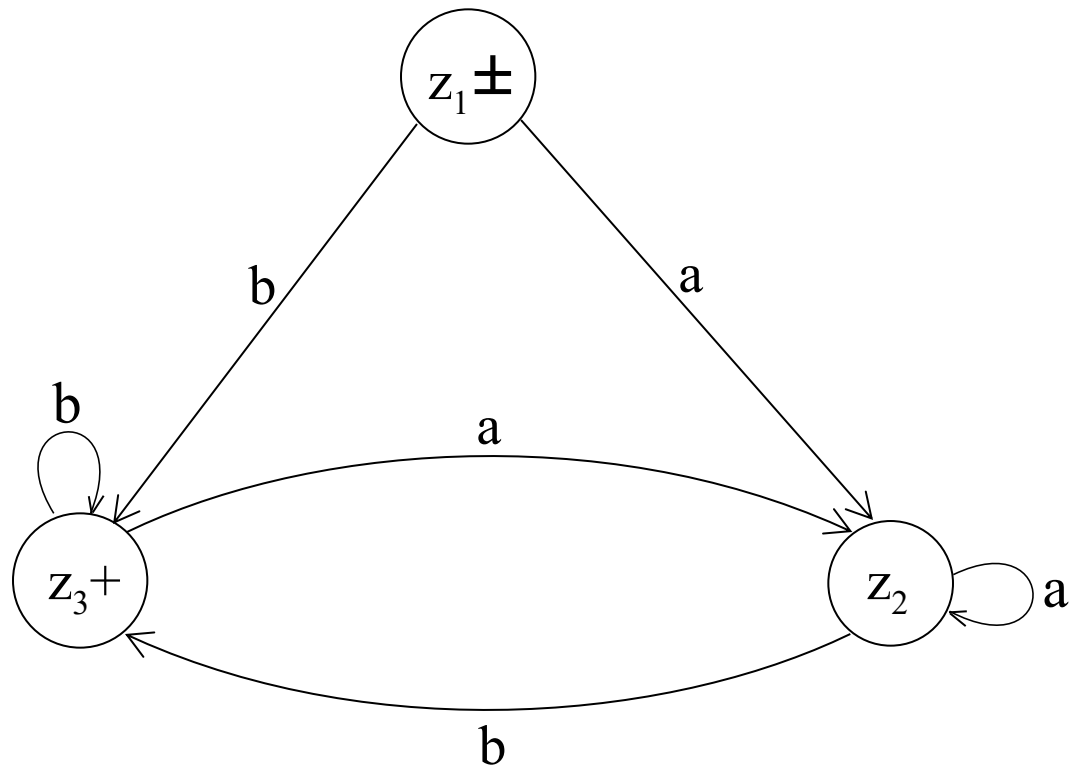
Old States	New States after reading	
	a	b
Final $z_1 \pm x_1$	$x_1 \pm z_2$	$(x_2, x_1) \pm z_3$

Example continued ...

Old States	New States after reading	
	a	b
Non-final $z_2 \sqcup x_1$	$x_1 \sqcup z_2$	$(x_2, x_1) \sqcup z_3$
$z_3^+ \sqcup (x_2, x_1)$	$x_1 \sqcup z_2$	$(x_2, x_1) \sqcup z_3$

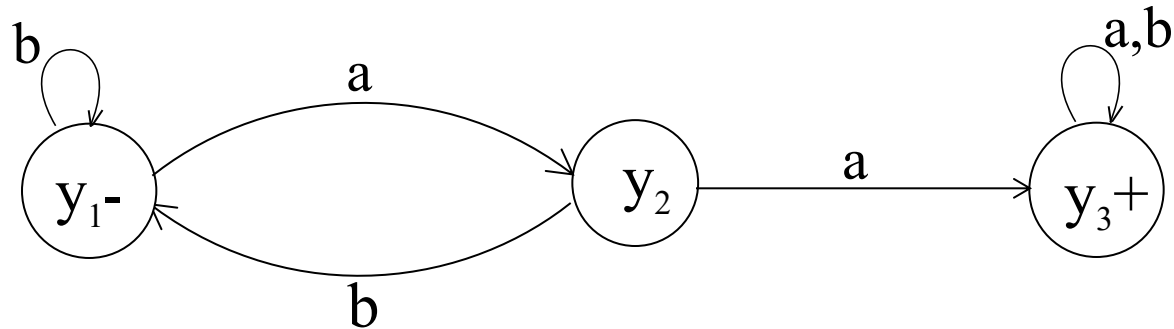
The corresponding transition diagram may be as under

Example continued ...

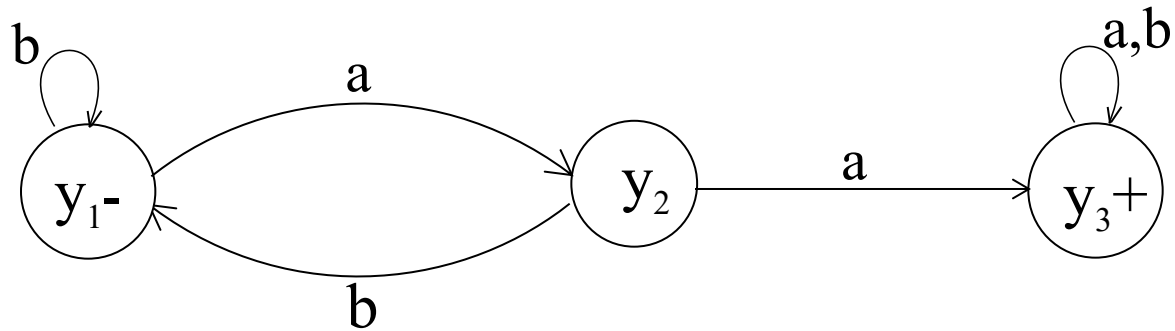
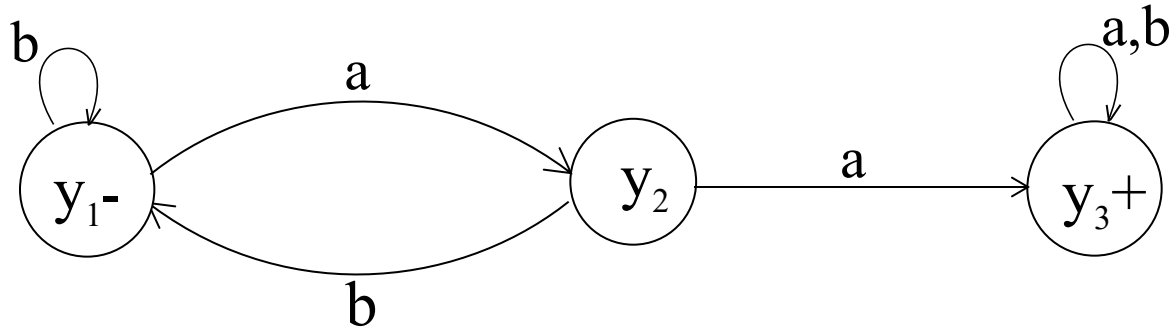


Example

Let $r=(a+b)^*aa(a+b)^*$ and the corresponding FA be



Example continued ...

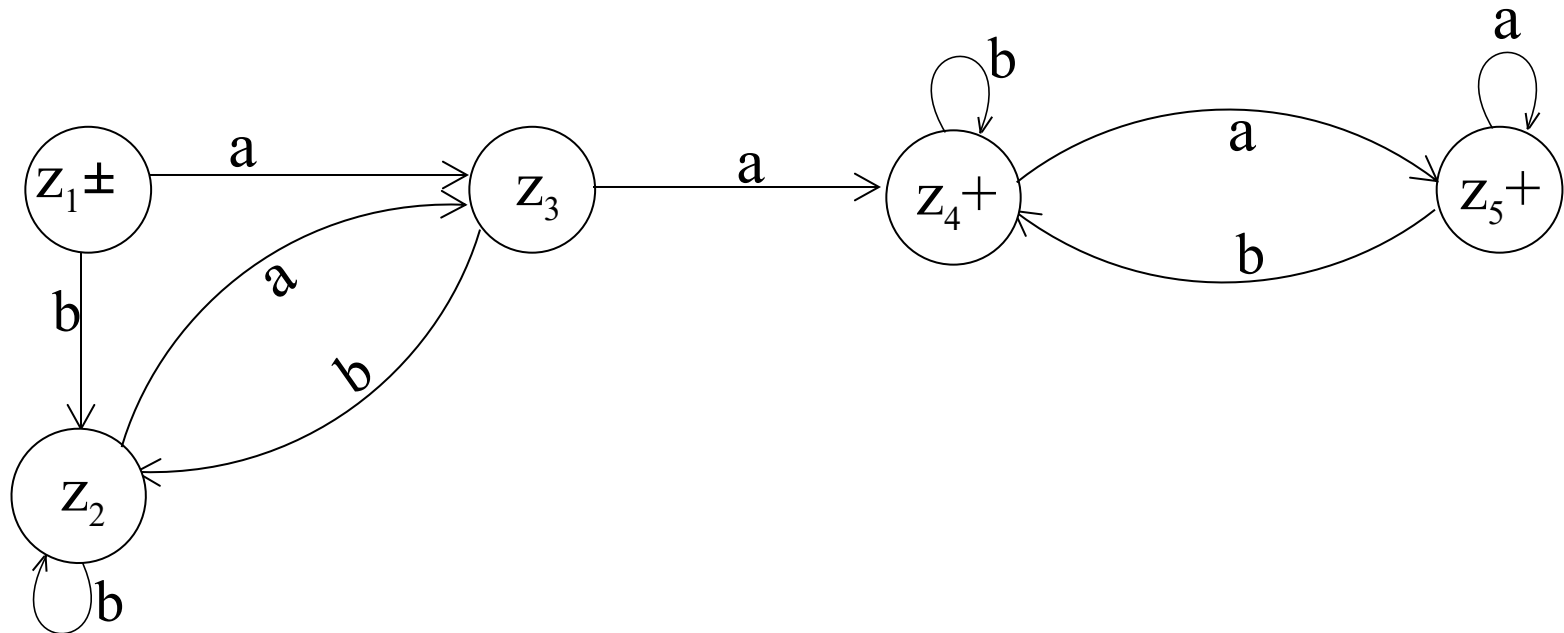


Old States	New States after reading	
	a	b
Final $z_1 \pm y_1$	$y_2 \pm z_3$	$y_1 \pm z_2$

Example continued ...

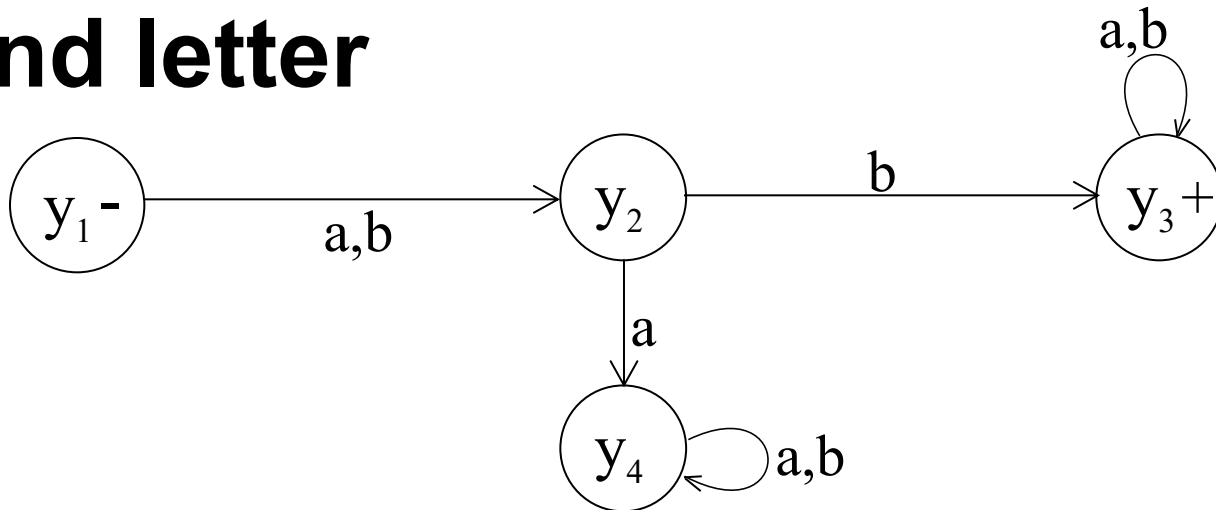
Old States	New States after reading	
	a	b
$z_2 \equiv y_1$	$y_2 \equiv z_3$	$y_1 \equiv z_2$
$z_3 \equiv y_2$	$(y_3, y_1) \equiv z_4$	$y_1 \equiv z_2$
$z_4^+ \equiv (y_3, y_1)$	$(y_3, y_1, y_2) \equiv z_5$	$(y_3, y_1) \equiv z_4$
$z_5^+ \equiv (y_3, y_1, y_2)$	$(y_3, y_1, y_2) \equiv z_5$	$(y_3, y_1) \equiv z_4$

Example continued ...

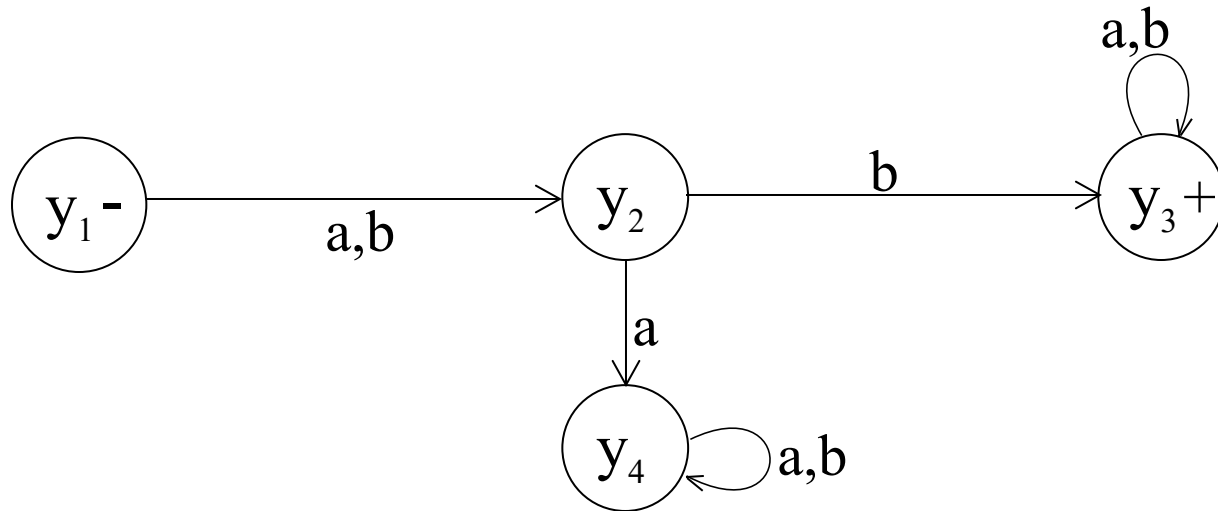


Example

Consider the following FA, accepting the language of strings with **b** as **second letter**



Example continued ...

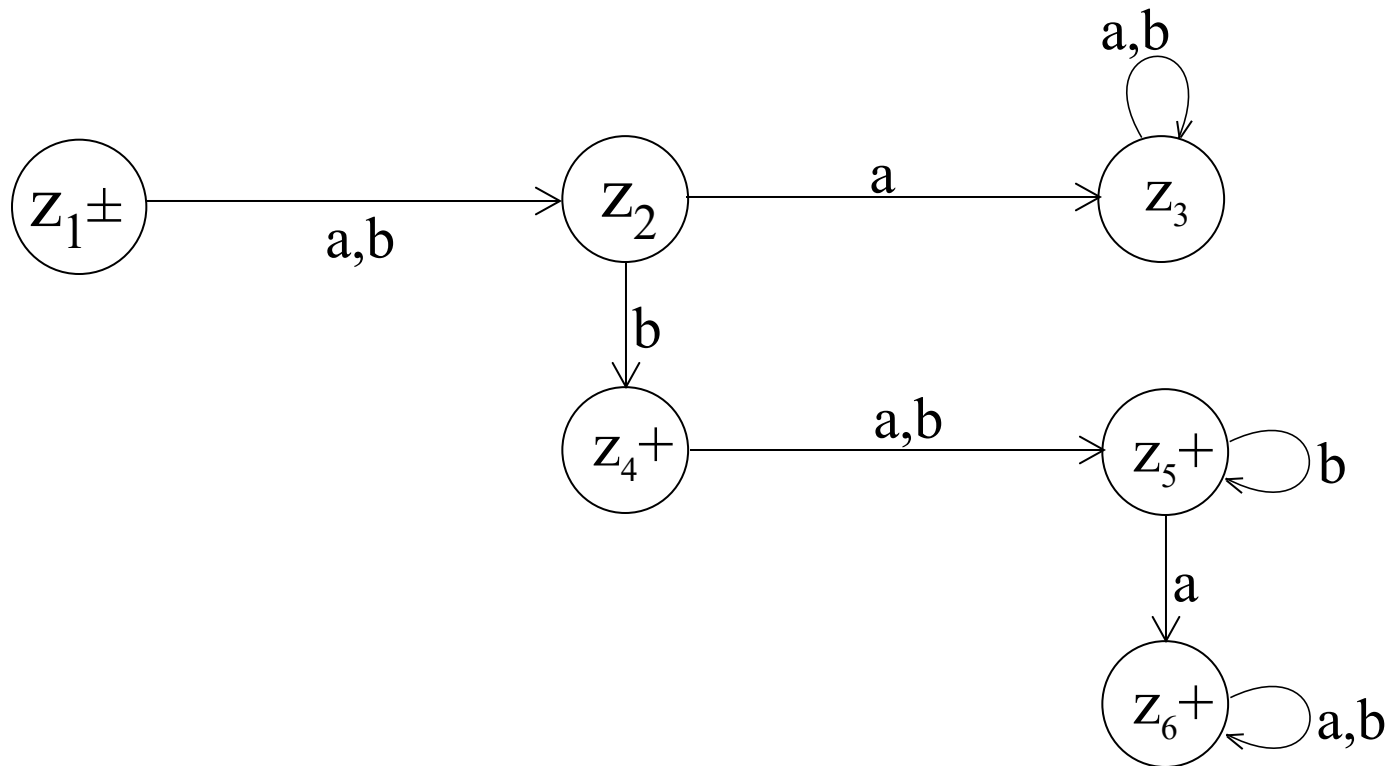


Old States	New States after reading	
	a	b
$z_1^{\pm}=y_1$	$y_2=z_2$	$y_2=z_2$

Example continued ...

Old States	New States after reading	
	a	b
$z_2 = y_2$	$y_4 = z_3$	$(y_3, y_1) = z_4$
$z_3 = y_4$	$y_4 = z_3$	$y_4 = z_3$
$z_4^+ = (y_3, y_1)$	$(y_3, y_1, y_2) = z_5$	$(y_3, y_1, y_2) = z_5$
$z_5^+ = (y_3, y_1, y_2)$	$(y_3, y_1, y_2, y_4) = z_6$	$(y_3, y_1, y_2) = z_5$
$z_6 = (y_1, y_1, y_2, y_4)$	$(y_1, y_1, y_2, y_4) = z_6$	$(y_1, y_1, y_2, y_4) = z_6$

Example continued ...



Summing Up

- Examples of Kleene's theorem part III (method 1) continued ,Kleene's theorem part III (method 2: Concatenation of FAs), Examples of Kleene's theorem part III(method 2:concatenation FAs) continued, Kleene's theorem part III (method 3:closure of an FA), examples of Kleene's theorem part III(method 3:Closure of an FA) continued